

WiP: Smartphone-based Acoustic Tissue Characterization for Personalized Body-contact Systems

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Abstract

Interactive systems such as haptic wristbands maintain sustained body contact, but variations in soft tissue mechanics can cause the same force to feel different across users. Conventional tissue characterization relies on specialized systems such as ultrasound elastography, limiting its use in personalized devices. We take a first step toward smartphone-based acoustic tissue characterization by investigating if a smartphone can estimate tissue relaxation time using its built-in speaker and microphone. The smartphone probes tissue with acoustic chirps and estimates relaxation behavior from tissue echoes. We validate the approach against laboratory indentation, observing a strong correlation between smartphone and mechanical estimates on a silicone phantom ($r = 0.96$) and a weaker positive correlation on the human forearm ($r = 0.29$). These preliminary results support the feasibility of smartphone-based acoustic tissue characterization for tissue-aware body-contact systems.

CCS Concepts

• Human-centered computing → Smartphones.

Keywords

acoustic sensing, mobile computing, tissue characterization, personalized body-contact systems

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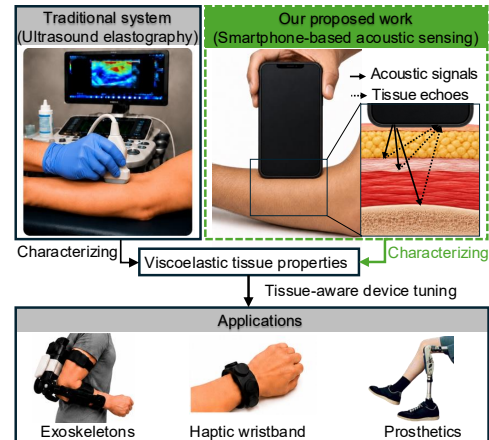


Figure 1: Our vision of acoustic tissue characterization using commodity smartphones.

1 Introduction

Interactive systems are increasingly designed to operate in sustained contact with the human body [5], including exoskeletons, haptic wristbands, and prosthetic sockets. The quality of these interactions depends on the user: the same strap tension, squeeze intensity, or socket pressure may feel different across individuals due to variations in soft tissue mechanics [5]. This variability motivates personalized body-contact systems that adapt their physical interaction to each user's tissue mechanics.

To characterize tissue mechanics, conventional approaches rely on specialized systems such as ultrasound elastography [13] or laboratory indentation rigs [10]. As illustrated in Fig. 1, these systems characterize viscoelastic tissue properties, including stiffness, viscosity, and relaxation time, which describe how soft tissue deforms, dissipates energy, and redistributes force under sustained loading. However, they require specialized clinical equipment, trained operators, or stationary laboratory testbeds, making them impractical for everyday deployment in personalized body-contact systems.

Our long-term vision is to make tissue characterization accessible using smartphones that people already carry. Smartphones are equipped with speakers and microphones, which

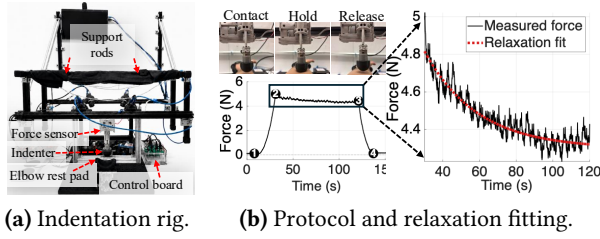


Figure 2: Indentation-based mechanical reference.

have been successfully used for acoustic sensing of gestures [1, 8, 12], respiration [7, 9, 14], eye blinks [4, 11], and, more recently, through-skin physiological signals [2, 6]. These advances suggest that smartphone acoustics can probe structures beneath the skin without specialized hardware. Thus, we ask a new question: *can commodity smartphone acoustics characterize the mechanical properties of soft tissue?*

Answering this question is challenging. Unlike clinical ultrasound systems, smartphone speakers and microphones provide coarse acoustic measurements with limited spatial resolution. Moreover, tissue responses are subtle and can be easily affected by motion and contact variation. Consequently, it remains unclear whether smartphone acoustics can recover meaningful viscoelastic tissue properties.

In this paper, we take a first step toward answering this question by investigating whether commodity smartphone acoustics can estimate a representative viscoelastic tissue property—relaxation time—using only the built-in speaker and microphone. Our approach actively probes tissue with high-frequency acoustic chirps, extracts tissue echoes, and estimates relaxation behavior from their temporal evolution.

We evaluate the proposed approach using a silicone phantom with controlled mechanical properties and an initial human pilot study. On the silicone phantom, smartphone-estimated relaxation time strongly correlates with the laboratory indentation reference ($r = 0.96$), while human forearm measurements show an initial positive trend ($r = 0.29$) under physiological motion and contact variability. These preliminary results provide initial evidence that commodity smartphone acoustics can capture viscoelastic tissue relaxation in controlled settings, suggesting a promising direction toward smartphone-based tissue characterization for personalized body-contact systems while highlighting robust on-body sensing as an important avenue for future work.

2 Mechanical Reference

We use an indentation rig as the mechanical reference for evaluating smartphone-based acoustic tissue characterization. As shown in Fig. 2, a displacement-controlled indenter presses a probe into the tissue while a load cell records the contact force. The standard stress-relaxation protocol consists of three stages: contact, hold, and release. During the

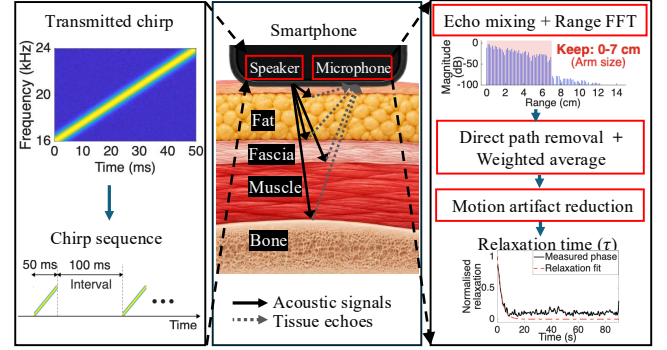


Figure 3: Overview of our proposed smartphone acoustic tissue characterization pipeline.

hold stage, the indentation depth remains fixed while the contact force gradually decays due to viscoelastic relaxation.

Following the standard stress-relaxation protocol, we fit the hold-phase force decay with a single-exponential model [3]:

$$F(t) = F_{\infty} + (F_0 - F_{\infty}) e^{-t/\tau}, \quad (1)$$

where F_0 and F_{∞} are the initial and steady-state forces, respectively, and τ is the relaxation time. We focus on relaxation time because it captures the rate at which tissue redistributes force under sustained loading, providing a compact descriptor of viscoelastic behavior. In this paper, the indentation-estimated relaxation time serves as the mechanical reference for evaluating smartphone-based estimates.

3 Acoustic Tissue Characterization

Fig. 3 summarizes our acoustic tissue characterization pipeline.

Acoustic Signal Configuration. The smartphone is configured to actively probe tissue using Frequency-Modulated Continuous Wave chirps. The bottom speaker repeatedly transmits chirps sweeping from 16–24 kHz over 50 ms, sampled at 48 kHz, with a 100 ms inter-chirp interval, while the bottom microphone records the returning tissue echoes.

Tissue Response Extraction. We first mix the transmitted and received chirp signals and compute a zero-padded 16,384-point range FFT to obtain a complex range profile. We retain the range bins within 7 cm, approximately the diameter of a human forearm, as the tissue-response region. To reduce direct-path leakage between the speaker and microphone, we subtract the first range bin from the remaining bins. Finally, we compute a weighted average of the complex coefficients, using their magnitudes as weights, to obtain a single complex tissue response for subsequent processing.

Motion Artifact Reduction. The amplitude and phase of the extracted response evolve with tissue relaxation but are also affected by involuntary arm or body motion. To reduce motion artifacts, we normalize the near-surface tissue response using a reference derived from distant range bins. These bins primarily capture common motion and residual

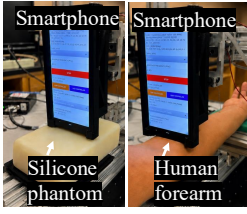


Figure 4: Smartphone measurement vs. indentation on the silicone phantom.

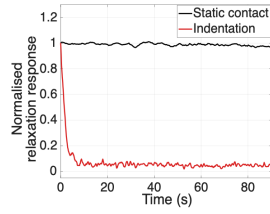


Figure 5: Static contact time estimation on the silicone phantom.

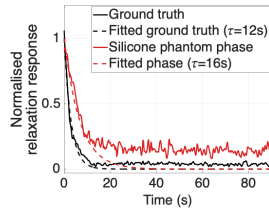


Figure 6: Relaxation time estimation on the silicone phantom.

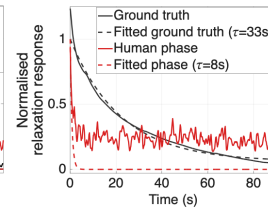


Figure 7: Relaxation time estimation on the human forearm.

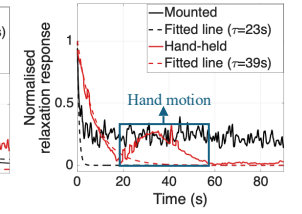


Figure 8: Mounted vs. hand-held smartphone measurements.

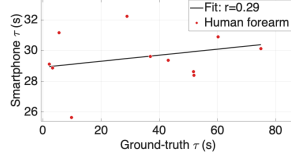
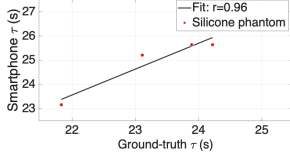


Figure 9: Correlation between smartphone-estimated and indentation-based relaxation times.

interference. This normalization reduces motion-induced variation while preserving the slower relaxation trend.

Relaxation-time Estimation. We use the phase of the normalized complex response because it is more sensitive than amplitude to the subtle tissue displacement. Assuming the phase follows the same relaxation behavior as the force in Eq. (1), we fit its temporal decay using the same exponential model to estimate the relaxation time, which is then compared with the indentation-based mechanical reference.

4 Implementation and Evaluation

Experimental Setup. We evaluate our acoustic tissue characterization against a laboratory indentation rig. As shown in Fig. 4, a Google Pixel 9 is mounted on the rig to collect acoustic measurements. The two modalities are synchronized using a three-cycle up-down motion detected by the smartphone’s IMU and the indentation rig. Experiments include 4 silicone phantom trials and 12 human forearm trials from 3 participants under an approved IRB protocol. We choose the forearm because it provides a relatively flat surface and represents a common attachment site for body-contact devices. The phantom provides a repeatable baseline, while human forearms introduce realistic motion and contact variability.

Validation of Tissue Response. Using a silicone phantom, we compare static smartphone-phantom contact with controlled indentation. As shown in Fig. 5, only the indentation condition exhibits a clear relaxation response, suggesting that the measured acoustic response is driven by tissue deformation rather than contact alone.

Relaxation-time Estimation. Fig. 6 and Fig. 7 compare representative smartphone-estimated relaxation responses with the mechanical reference for the silicone phantom and human forearm, respectively. The phantom closely follows the reference, while the human forearm preserves the overall

relaxation behavior but exhibits greater residual fluctuations due to physiological motion and contact variability.

Fig. 9 shows the estimated relaxation times across all trials. Smartphone estimates strongly correlate with the mechanical reference on the silicone phantom ($r = 0.96$), demonstrating the feasibility of acoustic tissue characterization in a controlled setting. On human forearms, the current implementation achieves a weaker but positive correlation ($r = 0.29$), indicating that improving robustness under realistic on-body conditions remains an important direction for future work.

Toward Hand-held Tissue Characterization. Fig. 8 compares mounted and hand-held measurements. Although hand motion introduces noticeable artifacts, the hand-held measurement still exhibits an overall relaxation trend. These results suggest the feasibility of hand-held tissue characterization while highlighting the need for more robust motion-artifact suppression in unconstrained use.

5 Limitations and Future Work

Robust on-body measurements. Our evaluation uses laboratory indentation as the mechanical reference. Hand motion and contact variability remain challenges, motivating improved motion compensation and contact-aware processing for reliable hand-held measurements.

Beyond relaxation time. We focus on relaxation time as a representative viscoelastic descriptor. Future work will estimate additional properties, such as stiffness and viscosity, for more comprehensive tissue characterization.

Generalization. Our evaluation uses one smartphone model, four silicone phantom trials, and twelve forearm measurements from three participants. Future work will evaluate more devices, body locations, and participants to assess cross-device, cross-location, and cross-user generalization.

Toward personalized body-contact systems. Future work will integrate smartphone-based tissue characterization into adaptive haptic wearables, exoskeletons, prosthetics, and other body-contact systems.

Acknowledgments

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